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10 – 14 March 2026, New Delhi

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On-Prem Agentic AI for Power & Utilities

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I am proud to have been recognized as a UiPath Most Valuable Professional from 2021 to 2026. I am also a LinkedIn Top Voice and a speaker at both national and international conferences. In addition, I am the creator of the YouTube channel "**Tutorials by Mukesh Kala.**"

Problem Statement

Power utility outage response is highly time-critical, but today it is fragmented, manual, and slow.

When an outage or equipment alarm happens, teams usually :

- look across multiple systems like SCADA, DMS, OMS, ERP, and SOP documents
- understand what the alarm means and how severe it is
- decide what action or restrictions are needed
- create or update a work order, often manually
- draft customer communication separately
- later reconstruct everything for audit, reporting, and handover

Real issue is not just the outage itself. The issue is that operators spend too much time gathering information, interpreting it, coordinating actions, and documenting the response across disconnected systems.

What are we demonstrating Today ?

- This use case demonstrates an offline, on-prem Agentic AI "Outage Response & Work Order Agent" for power utilities.

- When an outage or equipment alarm occurs, the agent can understand the incident, consult internal SOPs and past incidents, generate a structured ERP work order, and draft customer communications (including Indian languages all without sending any data to the internet.

- It is designed to act as a digital teammate that speeds up response and improves consistency, while keeping humans in control for high-risk actions.

What Happens Today ?

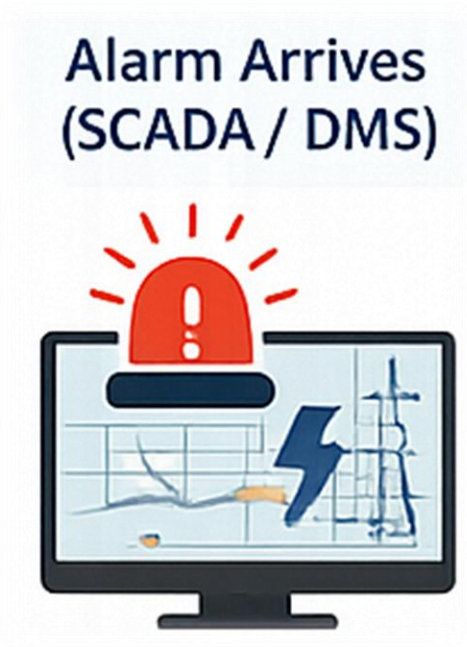
- **SCADA** = **Supervisory Control and Data Acquisition**
  Utility control system that monitors and controls field equipment in real time, such as substations, breakers, transformers, feeders, and alarms.

- **DMS** = **Distribution Management System**
  Operational system used by power distribution teams to manage the electricity network, analyze outages, isolate faults, reroute power, and support restoration decisions.

Scenario : A realistic incident

Incoming alarm (JSON payload)


```
{
  "ticket_id": "OUT-ALPHA-0007",
  "site": "Substation Alpha",
  "asset_id": "SS-ALPHA-FDR-12",
  "event": "Feeder trip alarm",
  "repeat_alarms": 2,
  "window_minutes": 10,
  "impact_customers": 450,
  "notes": "Crew Team-3 is nearby. Critical services impact is unknown."
}
```

Required outcomes in minutes:

- Correct severity and operational checklist
- ERP work order draft with priority and safety notes
- Customer communications using approved wording (English and Hindi)
- Traceable audit of sources used

Resource 1 : Outage Restoration SOP

This is the operations playbook that defines severity rules, safety requirements, and standard restoration steps.

Usually stored in:

- SharePoint / Confluence / Document Management System
- Internal QMS repository (ISO/utility compliance docs).
- Sometimes as PDFs maintained by Operations Excellence teams

```
# SOP: Outage Restoration & Switching Playbook (Synthetic)

## Safety First (mandatory)
- Confirm isolation and earthing before any field work.
- If transformer temperature alarm persists > 10 minutes, treat as high risk.
- Never recommend remote switching if protection status is unknown.

## Severity Classification
- SEV-1 (Critical): > 2000 customers OR critical services affected
(hospital/airport/water).
- SEV-2 (Major): 500-2000 customers OR repeated trip alarms within 30 minutes
OR substation equipment alarm.
- SEV-3 (Moderate): < 500 customers, single event, no critical services.
- SEV-4 (Minor): localized, no repeat alarms, no critical customers.

## Standard Restoration Steps
1. Validate outage via OMS/DMS and confirm affected feeders.
2. Check repeat trips/alarms and correlate with asset history.
3. If repeat alarms: dispatch crew and restrict switching changes until
clearance.
4. Prepare switching plan (recommendation only unless authorized).
5. Create ERP work order with correct priority and safety notes.
6. Customer communications: initial update + ETA, then periodic updates.

## Work Order Guidance (ERP/EAM)
- Priority: P1 for SEV-1, P2 for SEV-2, P3 otherwise
- Job Type: Fault Investigation for repeated trips
- Safety Notes: include isolation confirmation requirement
- Attach reference: outage ticket ID + feeder/substation identifiers
```

Resource 2 : Asset Register

This is how we understand what the asset is, where it is, and how critical it is. Helps severity, routing, and correct work order fields.

Usually stored in:

- EAM/ERP (SAP PM, Maximo, Oracle EBS)
- Asset Management system / GIS integration
- Exported nightly or accessed via internal API

	A	B	C	D	E	F
1	asset_id	asset_type	site	area	criticality	
2	SS-ALPHA-TX-01	Transformer	Substation Alpha	Indiranagar	High	
3	SS-ALPHA-FDR-12	Feeder	Substation Alpha	Indiranagar	High	
4	SS-BRAVO-FDR-07	Feeder	Substation Bravo	Whitefield	Medium	
5	SS-ALPHA-TX-02	Transformer	Substation Alpha	Indiranagar	High	
6	SS-ALPHA-FDR-13	Feeder	Substation Alpha	Indiranagar	High	
7	SS-ALPHA-FDR-14	Feeder	Substation Alpha	Indiranagar	Medium	
8	SS-ALPHA-CBR-03	Circuit Breaker	Substation Alpha	Indiranagar	High	
9	SS-ALPHA-RLY-02	Relay	Substation Alpha	Indiranagar	Medium	
10	SS-BRAVO-TX-02	Transformer	Substation Bravo	Whitefield	High	
11	SS-BRAVO-FDR-08	Feeder	Substation Bravo	Whitefield	Medium	
12	SS-BRAVO-FDR-09	Feeder	Substation Bravo	Whitefield	Medium	
13	SS-BRAVO-CBR-02	Circuit Breaker	Substation Bravo	Whitefield	High	
14	SS-BRAVO-RLY-04	Relay	Substation Bravo	Whitefield	Low	
15	SS-CHARLIE-TX-01	Transformer	Substation Charlie	Marathahalli	High	
16	SS-CHARLIE-FDR-05	Feeder	Substation Charlie	Marathahalli	Medium	
17	SS-CHARLIE-FDR-06	Feeder	Substation Charlie	Marathahalli	Medium	
18	SS-CHARLIE-CBR-01	Circuit Breaker	Substation Charlie	Marathahalli	High	
19	SS-CHARLIE-RLY-03	Relay	Substation Charlie	Marathahalli	Low	
20	SS-DELTA-TX-01	Transformer	Substation Delta	Electronic City	High	
21	SS-DELTA-FDR-03	Feeder	Substation Delta	Electronic City	High	
22	SS-DELTA-FDR-04	Feeder	Substation Delta	Electronic City	Medium	
23	SS-DELTA-CBR-02	Circuit Breaker	Substation Delta	Electronic City	High	
24	SS-DELTA-RLY-01	Relay	Substation Delta	Electronic City	Medium	
25	SS-ECHO-TX-01	Transformer	Substation Echo	HSR Layout	High	
26	SS-ECHO-FDR-02	Feeder	Substation Echo	HSR Layout	Medium	
27	SS-ECHO-FDR-03	Feeder	Substation Echo	HSR Layout	Low	
28	SS-ECHO-CBR-01	Circuit Breaker	Substation Echo	HSR Layout	Medium	
29	SS-FOXTROT-TX-01	Transformer	Substation Foxtrot	Koramangala	High	
30	SS-FOXTROT-FDR-01	Feeder	Substation Foxtrot	Koramangala	Medium	
31	SS-FOXTROT-FDR-02	Feeder	Substation Foxtrot	Koramangala	Medium	
32	SS-FOXTROT-CBR-01	Circuit Breaker	Substation Foxtrot	Koramangala	High	

Sheet1

Resource 3 : Past Incident Library (Knowledge)

This is the experience memory : common patterns, fixes, and lessons learned. Used to suggest more practical next steps ("this looks like the last incident…").

Usually stored in:

- Incident history exports
- ITSM tickets (ServiceNow/Jira)
- Post-incident review documents
- Reliability engineering knowledge base


```
[
  {
    "incident_id": "INC-2025-1120",
    "summary": "Repeat feeder trip on SS-ALPHA-FDR-12, 3 times in 20
minutes",
    "impact_customers": 860,
    "actions_taken": ["Crew dispatched", "ERP work order created", "Switching
restricted pending clearance"],
    "resolution": "Loose termination identified and repaired",
    "lessons": "Repeat trip pattern correlated with recent maintenance; add
post-maintenance inspection checklist."
  },

{
    "incident_id": "INC-2025-0902",
    "summary": "Transformer temperature alarm SS-ALPHA-TX-01 with rising
trend",
    "impact_customers": 1200,
    "actions_taken": ["Load redistribution recommended", "Cooling inspection
work order raised"],
    "resolution": "Cooling fan relay replaced",
    "lessons": "Lowered temperature trend threshold improved early response."
  }
]
```

Resource 4 : Customer Communication Templates

Utilities usually have approved wording for customer messages to avoid panic, reduce complaints, and stay compliant.
Templates to draft consistent comms.

Usually from :

- Corporate communications templates
- CRM knowledge base
- Call-center scripts repository

```
# Customer Message Templates

## SMS - English
Power supply is temporarily interrupted in {area}. Our teams are working to
restore service. Estimated restoration: {eta}. Ref: {ticket_id}


## SMS - Hindi
{area} में बिजली आपूर्ति अस्थायी रूप से बाधित है। हमारी टीम सेवा बहाल करने के लिए काम कर रही
है। अनुमानित समय: {eta}. संदर्भ: {ticket_id}


## Email - English
Subject: Power Interruption Update - {area} (Ref: {ticket_id})
Body:
Dear Customer,
We are currently responding to a power interruption affecting {area}. Our
teams are working to restore service.
Estimated restoration time: {eta}.
We will share updates if timelines change.
Regards,
Utility Operations
```

Resource 5 : Alarm / Incident Intake (Transactional)

Utilities usually have approved wording for customer messages to avoid panic, reduce complaints, and stay compliant.
Templates to draft consistent comms.

Usually from :

- Corporate communications templates
- CRM knowledge base
- Call-center scripts repository

```
# Customer Message Templates

## SMS - English
Power supply is temporarily interrupted in {area}. Our teams are working to
restore service. Estimated restoration: {eta}. Ref: {ticket_id}


## SMS - Hindi
{area} में बिजली आपूर्ति अस्थायी रूप से बाधित है। हमारी टीम सेवा बहाल करने के लिए काम कर रही
है। अनुमानित समय: {eta}. संदर्भ: {ticket_id}


## Email - English
Subject: Power Interruption Update - {area} (Ref: {ticket_id})
Body:
Dear Customer,
We are currently responding to a power interruption affecting {area}. Our
teams are working to restore service.
Estimated restoration time: {eta}.
We will share updates if timelines change.
Regards,
Utility Operations
```

What is the Problem ?

Operators coordinate decisions, actions, and communication under pressure

Challenge

Minutes matter

Information

Siloed

Execution

Manual

Impact

- Slower response and longer outages
- Inconsistent actions between operators
- Higher operational risk (missed safety steps)
- Customer dissatisfaction during incidents
- Limited traceability for governance

Utility problem: outage response is time-critical and fragmented

Proposed solution: on-prem agentic workflow

Local model, internal knowledge, secure tools

Event Intake (SCADA / OMS) → **On-prem Agent (Brain + Memory)** → **Secure Tools (APIs / Connectors)**

Internal knowledge stays on-prem

- SOPs and runbooks
- Asset register and criticality
- Past incidents and lessons learned
- Approved customer templates
- Optional Indian language glossary

Agent produces action-ready package

- Severity classification
- Operational checklist
- ERP work order draft (structured)
- Customer communications (English and Hindi)
- Audit trail and assumptions

Let’s see the Agent in Action

What the demo output signifies

Each section maps to a real operational need

Incident summary

- Understands the alarm payload
- Classifies severity using SOP logic
- Explains why in one line

Recommended actions

- Operator-ready checklist
- Ordered steps (validate, dispatch, restrict, log)
- Consistent across shifts

ERP work order draft

- Structured data ready for API integration
- Priority aligns to severity
- Safety notes and reference included

Customer comms and audit trail

- Template-driven messaging
- Supports Indian languages
- Traceability: SOP, templates, incident history cited

Let's Start Simple

1. Manual Car

This is the starting point, where everything depends on human effort.

Example: A child plays with a manual toy car, pushing it around using their hand. If the car gets stuck, the child has to physically move it around the obstacle. The car cannot do anything on its own.

Characteristics:

- Entirely dependent on the user for movement.
- No automation or intelligence every action is manual.
- If the child doesn’t push it, the car does nothing.

In the context of automation:

The manual car represents **No automation at all** every action in a task is done by a human.

Let's Start Simple

2. Remote-Controlled Car (RPA):
This is the first step toward automation tasks are easier but still require explicit instructions.

Example: Now, the child gets a remote-controlled car. The car moves forward, backward, and turns based on buttons pressed on the remote control. The child doesn't need to physically push the car anymore, but they still need to guide it.

Characteristics:
- The car follows commands given via the remote.
- It performs repetitive actions but cannot think or adapt.
- If an obstacle appears, the child must manually steer it around the obstacle using the remote.

In the context of automation:
The remote-controlled car represents **RPA (Robotic Process Automation).**
RPA automates repetitive tasks but relies entirely on pre-programmed instructions and cannot handle anything outside of those instructions.

Example in the real world: An RPA bot processing invoices by extracting data from fixed formats.

Let's Start Simple

3. Autonomous Car (Agentic AI):
This is the leap to intelligence and independence—no manual intervention is required.

Example: The child now has an autonomous car. The car has sensors, a camera, and intelligence built in. If you tell the car, “Drive to the other side of the room,” it:

- **Perceives** the environment (using its sensors to detect obstacles).
- **Plans** the best route to reach the other side.
- **Acts** by driving itself, adjusting the route in real-time if something unexpected appears (like a toy in its path).

Characteristics:

- The car works independently based on its understanding of the environment.
- It adapts to changes and solves problems without needing explicit instructions.
- It is goal-oriented: If its path is blocked, it recalculates and continues toward the goal.

In the context of automation:
The autonomous car represents **Agentic AI.**
Agentic AI can perceive, plan, and act autonomously, making it capable of handling complex and dynamic scenarios.

Example in the real world: An agentic AI system in business might handle customer complaints by analyzing emails, deciding the best response, and executing the required actions—without needing step-by-step guidance.

Take Away

Manual Car	Remote-Controlled Car (RPA)	Autonomous Car (Agentic AI)
Entirely human-driven - no automation at all. Everything dependent on Human and No Automation	Automates repetitive actions but requires explicit control and guidance. Actions are predefined	Fully independent, capable of perceiving, planning, and acting to achieve goals without constant human intervention.

Who will Fetch the Ball ?

Imagine you're playing in a park with a ball, and you accidentally kick it into the bushes.

Who will Fetch the Ball ?

Instead of you having to go fetch it, there's a helpful robot called **"Scout"**

How Scout (Agentic AI) Works:

1. **Understanding:** Scout notices that the ball is missing and sees where it went.
2. **Planning:** It thinks, "The ball is in the bushes. I need to go around the bench and carefully pick it up."
3. **Action:** Scout rolls over, avoids obstacles like a puddle and a rock, grabs the ball, and brings it back to you.

You didn't need to tell Scout step-by-step what to do. You simply said, "Scout, the ball is gone!" and it figured out everything else.

Definitions

What is an Agent?

- An AI-powered entity that can follow instructions, use tools, run automations or make decisions based on human language defined goals.
- It interacts with its environment, collect information, analyze data and self-determine tasks to meet the given goals.
- Agents can be used to automate complex tasks that require language understanding and decision-making.

Agent key principles

- **Planning -** Understand a given set of goals and plan actions to achieve them.
- **Tool usage -** Make use various tools, such as RPA automations or Integration Service connectors, to perform tasks effectively.
- **Context -** Detect and filter hallucinations in LLM responses when a reference source and a user query is provided.
- **Adaptability -** Perceive and process information from their environment, to make them more interactive and context aware.
- **Reflection & Memory -** Remember interactions (tool usage and perception), behaviors (tool usage and planning) and critique its past actions and regenerate responses.

How agents are built: Brain, Memory, Tools

A simple architecture you can map to your current stack

Brain

- Local LLM/SLM for reasoning
- Planning and structured outputs
- Policy-aware decisions

Memory

- Short-term state (this incident)
- Long-term knowledge (SOPs, history)
- Feedback loop (learn from outcomes)

Tools

- Secure APIs to ERP/CRM
- Create work orders and tickets
- Send communications with approvals

Key principle: tools are permissioned and logged. Agents do not free-run.

Benefits for utilities

Operational, customer, and governance value

Operational efficiency

- Faster response during incidents
- Less manual effort for work orders and updates
- Consistent execution across teams

Reliability and safety

- SOP-driven checklists reduce missed steps
- Approvals can be enforced for high-severity actions
- Clear escalation and handover

Customer experience

- Proactive and consistent communications
- Reduced call-center load during outages
- Localized messaging

Governance and security

- On-prem deployment for sensitive data
- Audit trail of sources used and assumptions
- Role-based controls and policy enforcement

Why on-prem matters (and how to keep it safe)

Data residency, resilience, and controlled actions

On-prem design principles

- Internal documents stay inside the utility network
- Local vector database (RAG) instead of uploading data externally
- Model runtime hosted locally (LLM or SLM)
- No dependency on external connectivity for incident response

Responsible AI controls

- Tool permissions (what the agent is allowed to do)
- Human-in-the-loop approvals for high-severity actions
- Structured outputs only (no uncontrolled execution)
- Logs: inputs, sources used, actions generated

What else can be done next

A practical expansion path for agent adoption

Near-term extensions (0–3 months)

- Connectors to ERP, CRM, OMS, GIS via secure APIs
- Approval gates for switching and public notifications
- Critical customer detection from GIS layers
- Shift handover summaries and incident reporting

Broader use cases (3–12 months)

- Predictive maintenance triage and work order automation
- Billing and collections: dispute triage and outreach
- Data delay chaser agent for data timeliness
- Field crew scheduling support with constraints
- Multilingual self-service customer support

Key takeaway

- Agentic AI enables utilities to move from AI that answers to AI that executes safely.
- On-prem architectures make it feasible to use confidential operational data while maintaining governance.

THANK YOU

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THANK YOU

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